Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **distributed Python/C++ systems**, cloud services, automated validation, observability, and **reliable high-throughput platforms** across production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production observability supporting dependable distributed workflows reliably at scale.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions and failure isolation, preserving recoverable background processing under sustained production workloads reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Processed 1K+ files per batch using **multithreaded Python services**, validating heterogeneous inputs while isolating failures and preserving responsive workflows during concurrent model execution and sustained production traffic reliably.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured automated testing**, model switching, fallback paths, persistent results, and reviewer controls supporting reproducible production workflows and systematic debugging.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **reproducible evaluation pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and controlled experiments across diverse scenarios reliably.
- Improved closed-loop planning stability by 15% through a **simulation-based validation model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex scenarios reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments under sustained traffic.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Pandit Deendayal Energy University | *Research Fellowship* | *India* **January 2023 – May 2024**
- Designed a **distributed authentication architecture** spanning users, IoT devices, and role-based access scenarios, evaluating immutable identity records and self-sovereign credential models to reduce reliance on centralized identity stores.
- Published peer-reviewed security research after analyzing protections against replay, phishing, man-in-the-middle, and credential-theft attacks, documenting access-control trade-offs and presenting technically defensible decentralized authentication designs at IEEE conferences.

Skills

Languages: C++, Python, Java, SQL, JavaScript, TypeScript, Bash
Systems & Design: Data Structures, Algorithms, Distributed Systems, Systems Programming, Object-Oriented Programming, Multithreading, Concurrent Programming, Operating Systems, Linux, System Design
Cloud & Data: AWS, Azure, GCP, Cloud Storage, PostgreSQL, MySQL, MongoDB, AWS RDS, Data Modeling, Caching
Reliability & Quality: Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Debugging, Observability, Monitoring, Error Handling, Fault Isolation, Performance Optimization, Production Reliability
Delivery & Tools: Git, GitHub, Jenkins, Docker, Kubernetes, Maven, CI/CD, Build Automation, Version Control
Engineering: Microservices, REST APIs, JSON, Simulation, Requirements Analysis, Technical Documentation, Code Review, Software Development Lifecycle

Projects

Multi-Cloud Cluster Reliability Harness - Design Concept
- Designed a **C++/Python cloud-cluster harness concept** provisioning isolated workloads, injecting failures, measuring latency and recovery, validating idempotent operations, and exporting observability signals across AWS, Azure, and GCP scenarios.

Distributed Storage Diagnostic Toolkit - Design Concept
- Designed a **distributed-storage diagnostic toolkit concept** generating configurable I/O workloads, tracing request lifecycles, detecting performance regressions, validating replication behavior, and visualizing throughput, latency, and failure-recovery results across Linux environments.

CPU-Scheduling Simulator | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.